

NumPy Complete Placement Cheat Sheet (Interview Edition)

A compact handbook covering the core NumPy API, interview differences, memory tricks, and placement-focused notes.

1. Import

```
import numpy as np
```

2. Why NumPy?

Fast (C backend), vectorized, memory efficient, broadcasting, multidimensional arrays.

3. Array Creation

Method	Purpose
array()	Create array from list/tuple
zeros()	Array filled with 0
ones()	Array filled with 1
empty()	Uninitialized values
full()	Fill with constant
eye()	Identity-like matrix
identity()	Identity matrix
arange()	Range with step
linspace()	Equal spacing
logspace()	Logarithmic spacing
diag()	Diagonal matrix
asarray()	Convert to ndarray
copy()	Independent copy

4. Array Properties

Property	Meaning
shape	Dimensions
ndim	No. of dimensions
size	Total elements
dtype	Data type
itemsize	Bytes per element
nbytes	Total bytes
T	Transpose

5. Indexing & Slicing

`arr[0], arr[-1], arr[1:4], arr[::-1], arr[:,1], arr[1,:], arr[[0,2]], arr[arr>5],
np.where(cond,x,y)`

6. Views & Copies

Function	Meaning
<code>view()</code>	Shares memory
<code>copy()</code>	Independent memory
<code>flatten()</code>	1D copy
<code>ravel()</code>	1D view if possible

7. Broadcasting

Rules: Compare shapes right→left. Dimensions must match or one must be 1.
Broadcasting stretches; it does not copy.

8. Universal Functions

Function	Purpose
add subtract multiply divide	Arithmetic
power mod remainder	Power & remainder
sqrt square abs	Basic math
exp log log10	Exponential & logs
sin cos tan	Trigonometry
maximum minimum	Elementwise comparison
logical_and/or/not	Logical operations

9. Aggregation

Method	Purpose
sum	Total
prod	Product
mean	Average
median	Middle value
std	Std deviation
var	Variance
min/max	Min/Max value
argmin/argmax	Index of min/max
percentile	Percentile
any/all	Boolean aggregation

10. Axis

axis=0 → Columns

axis=1 → Rows

11. Reshaping

Method	Purpose
reshape()	Change shape
reshape(-1)	Auto dimension
resize()	Modify original
transpose()/T	Swap rows & columns
expand_dims()	Add dimension
squeeze()	Remove size-1 dimension

12. Joining

Method	Purpose
concatenate	Join arrays
stack	Join with new axis
vstack	Vertical
hstack	Horizontal
column_stack	Columns
row_stack	Rows

13. Splitting

Method	Purpose
split	Equal split
array_split	Unequal split
vsplit	Vertical split
hsplit	Horizontal split
dsplit	Depth split

14. Random

Method	Purpose
rand	Uniform 0-1
randn	Normal distribution
randint	Random integers
choice	Random selection
shuffle	Shuffle in-place
permutation	Shuffled copy
seed	Repeatability

15. Sorting & Searching

Method	Purpose
sort	Sort values
argsort	Sort indices
unique	Unique values
where	Conditional
nonzero	Non-zero indices
isin	Membership test

16. Linear Algebra

Method	Purpose
dot	Dot product
matmul/@	Matrix multiplication
inner	Inner product
outer	Outer product
linalg.inv	Inverse
linalg.det	Determinant

17. Statistics

mean, median, std, var, corrcoef(), cov()

18. Interview Differences

Concept	Difference
shape vs size	Dimensions vs total elements
ndim vs shape	Count vs tuple
view vs copy	Shared vs independent
flatten vs ravel	Copy vs view
max vs argmax	Value vs index
concatenate vs stack	Join vs new axis
vstack vs hstack	Vertical vs horizontal
rand vs randn	Uniform vs normal
Boolean indexing vs where	Filter vs replace

19. Memory Tricks

ARG=index
flatten=copy
ravel=view
axis0=columns
axis1=rows
Broadcast=stretch
reshape(-1)=NumPy calculates

20. Placement Patterns

Filtering, aggregation, broadcasting, reshape before ML, matrix multiplication, random dataset generation, boolean masking.